LED feedback and timestamp consistency

Objective

Detect a breadboard LED flash with CAM2. Send one timing event after detection. Check that both receivers record the event and agree on its timestamp.

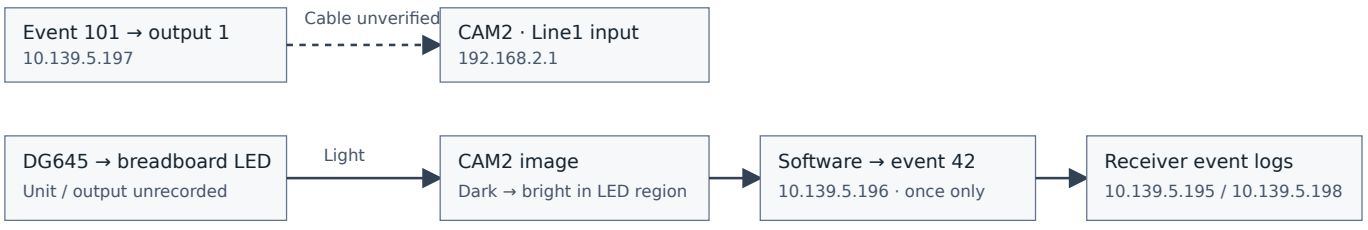
Status: LED feedback is configuration 5 and the primary test. It has not run. The completed sequence and camera tests below provide reference measurements.

Setup

Equipment

Device	IP address	PV prefix and function
Event generator	10.139.5.196	`EVT:DEV2:20:`; sends events 101 and 42
Receiver A	10.139.5.195	`EVT:DEV2:10:`; records event timestamps
Receiver B	10.139.5.198	`EVT:DEV2:40:`; records event timestamps
Output receiver	10.139.5.197	`EVT:DEV2:30:`; output 1 selects event 101
CAM2	192.168.2.1	`CAM:DEV2:20:`; external Line1 trigger; 1 ms exposure; 1200 × 900 Mono12
DG645 and LED	Connected unit unrecorded	Breadboard LED connects to a DG645; unit and output require identification
Recording host	10.139.4.59	Runs the IOC and analysis; camera interface: 192.168.2.100

The timing chassis use firmware and IOC release 20261017. The clock frequency is 125 MHz. Receiver timestamps have 8 ns resolution.



Event 101 is not yet verified as the cause of the LED flash. Image detection starts the software response.

Image detection starts the response event. The LED drive path still requires verification.

The camera-trigger cable and DG645 output have not been identified. Do not assume that event 101 causes the LED flash. A photodiode connected to a verified timing input is required to measure the optical edge with hardware timestamps.

Procedure

1. Identify the LED drive cable and camera trigger cable. Record both endpoints. Keep the existing camera acquisition and DG645 settings.
2. Read receiver status. Require valid links and timestamps, with no FIFO overflow. Save the current controls and event maps.
3. Review the exact commands before the live trial. Use configuration 5: event 101 at 1 Hz; output delay 0 ns; pulse width 100,000 ns. Load event 42 at 0 ns and stop code 255 at 1,000 ns in sequencer bank 1.
4. Record for 30 seconds. Save receiver logs, camera metadata and image arrays. Use image columns 300–699 and rows 550–849, with zero-based coordinates.
5. Wait for a new frame with a region peak of 1,000 counts or less. Then wait for a new frame with a peak above 1,000 counts.
6. After that transition, send event 42 once. Check for one receipt on each receiver. Reserve the final three seconds for receiver readout. If no transition occurs, send no response event.
7. Disarm the sequencer. Restore the saved controls and event maps. Read them again to confirm restoration.

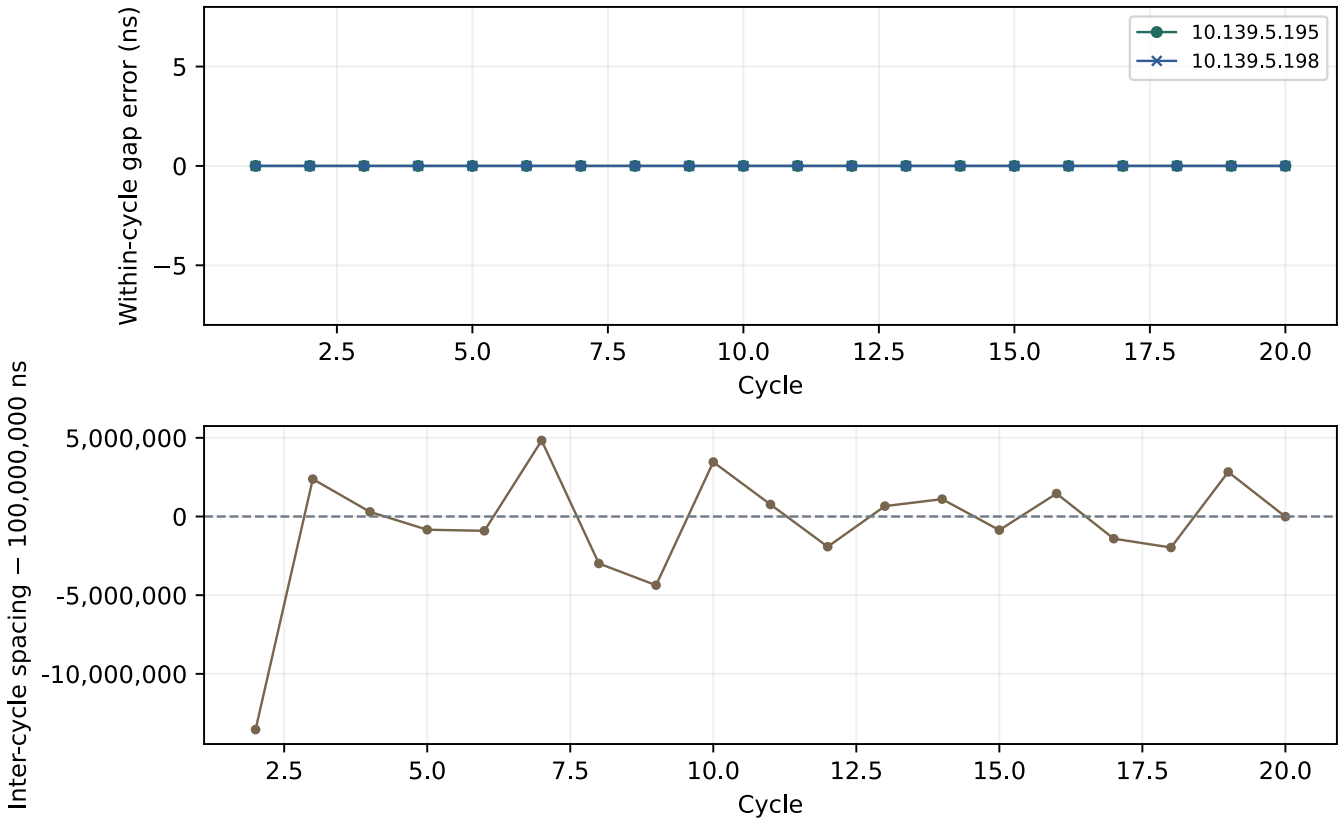
Record the dark and bright frame IDs, region peaks, response event timestamps and receiver difference. Record the host time from image detection to sequence confirmation separately. This host interval includes software, network and readback delays. It is not an optical delay measurement.

Results

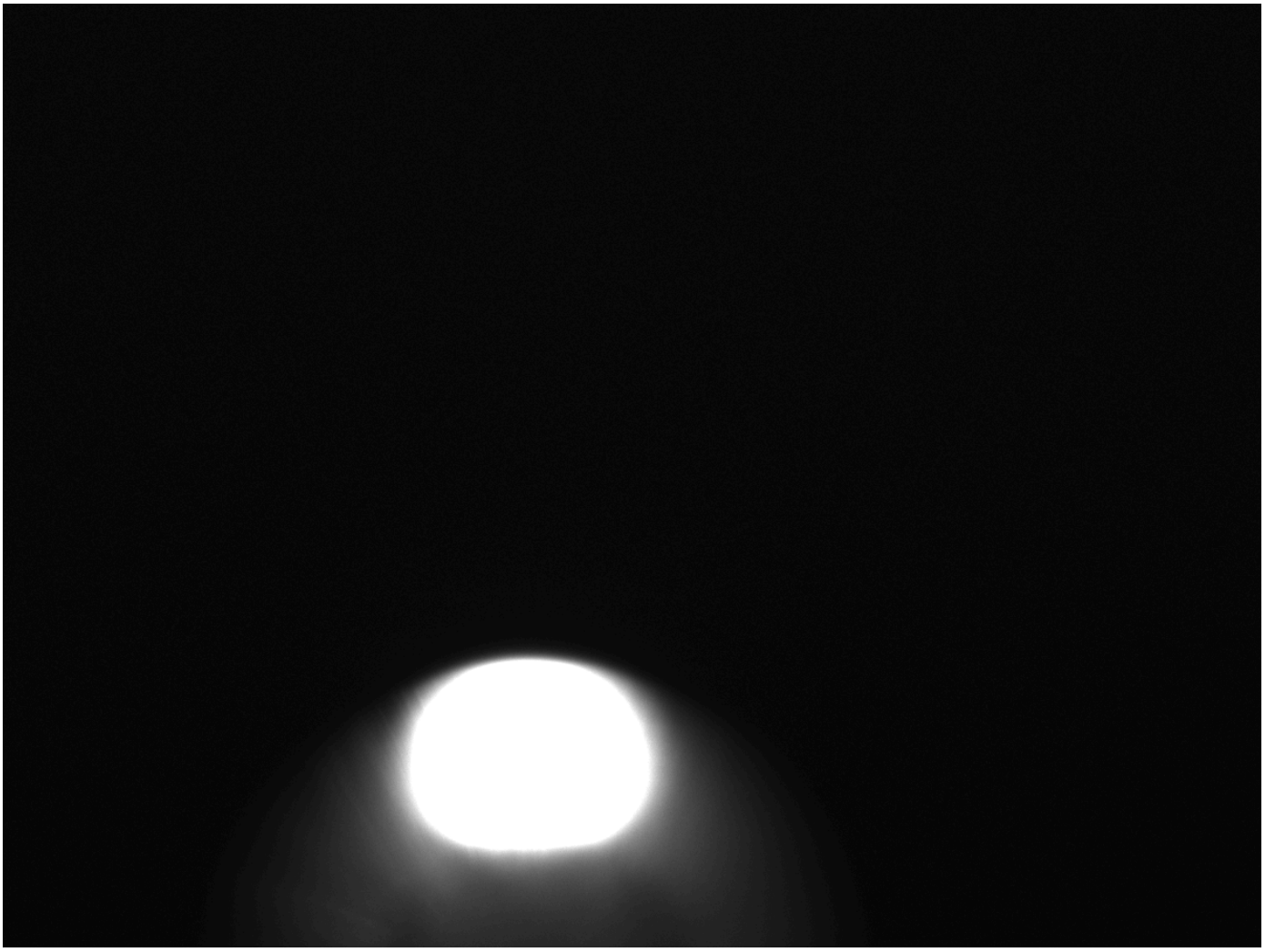
The LED-feedback test has no measured result yet. The reference tests were recorded on November 3–4, 2026. The starting settings were restored and verified after each changed test.

Reference measurements

Test	Received	Timing result
Single event	1 of 1 on each receiver	Receiver difference: 0 ns
Two events	2 of 2 on each receiver	Gap: 1,000 ns; receiver difference: 0 ns
Multiple delays	3 of 3 on each receiver	Gaps: 1,000 ns and 9,000 ns; receiver difference: 0 ns
Repeated sequence	40 of 40 on each receiver; 20 cycles	Each pair: 1,000 ns; gap-error standard deviation: 0 ns
Previous camera-rate test	60 images at 1 Hz; no frame-ID gaps or save overruns	58 receiver pairs: 0 ns difference; 0 of 60 LED-bright images



All 20 event pairs retained the programmed gap. Host pacing varied between cycles.



10 Hz reference: LED bright.



1 Hz trial: no bright LED detected.

See the test records for exact UTC timestamps, settings, commands and data files. The previous camera-rate test did not send a response event after image detection.

Findings

Both receivers recorded all expected sequence events. Their timestamps agreed at 8 ns resolution. These measurements support event delivery and timestamp consistency for CXFEL at the tested settings. They do not establish electrical output delay, absolute UTC accuracy or 1 kHz operation.

The camera received frames at 1 Hz, but the images did not establish LED-flash capture. Verify the LED drive and exposure timing before the feedback trial. Camera-based feedback can test the complete software response. Hardware optical timing requires a photodiode input.

▸ **Full test records**

▸ **LED feedback commands**